



# Weather Information using OpenWeatherMap API



## Objective

This project demonstrates how to retrieve real-time weather information using the **OpenWeatherMap API**. Users can enter a city name to view current weather conditions including temperature, humidity, wind speed, and weather description.

## 🔧 Technologies Used

-  Python
-  Requests Library
-  OpenWeatherMap API
-  Pandas

```
In [1]: import requests
import pandas as pd
```

```
In [2]: API_KEY = "09347103040aaebdd7d8da1b44bc26c4"
```

```
In [10]: city = input("Enter City Name: ")

url = f"https://api.openweathermap.org/data/2.5/weather?q={city}&appid={API_KEY}&units=metric"

response = requests.get(url)

data = response.json()
```

```
In [11]: weather_data = []

if response.status_code == 200:

    weather_data.append({
        "City": data["name"],
        "Country": data["sys"]["country"],
        "Temperature (°C)": data["main"]["temp"],
        "Feels Like (°C)": data["main"]["feels_like"],
        "Humidity (%)": data["main"]["humidity"],
        "Weather": data["weather"][0]["main"],
        "Description": data["weather"][0]["description"],
        "Wind Speed (m/s)": data["wind"]["speed"]
    })

else:

    print("Error:", data.get("message"))
```

```
In [12]: df = pd.DataFrame(weather_data)

df
```

```
Out[12]:
```

|   | City      | Country | Temperature (°C) | Feels Like (°C) | Humidity (%) | Weather | Description | Wind Speed (m/s) |
|---|-----------|---------|------------------|-----------------|--------------|---------|-------------|------------------|
| 0 | Australia | AU      | 17.21            | 16.14           | 44           | Clear   | clear sky   | 7.49             |

```
In [13]: df.to_csv("weather_information.csv", index=False)

print("CSV File Saved Successfully.")
```

CSV File Saved Successfully.



## Conclusion

Successfully retrieved real-time weather information using the **OpenWeatherMap API**. The application allows users to search weather details by city name.

- 🏠 City & Country
- 🌡️ Temperature
- 😊 Feels Like Temperature
- 💧 Humidity
- ☁️ Weather Condition
- 📄 Description
- 🌀 Wind Speed

The retrieved data was organized into a **Pandas DataFrame** and exported as a **CSV file** for future analysis and reporting.

Weather API Project Completed Successfully!